

Lab Experiment-V: Working with flow meters

Objectives

- (i) Get hands-on experience with various types of flow meters used in industrial and laboratory applications.
- (ii) Learn about the working principles of various flow meters.
- (iii) Quantification of characteristic constants with analog flow meters (such as venturimeter, orificemeter, etc.)

The Essential Background

1. Read up on the working principles of (i) an Orificemeter, (ii) a Venturimeter, (iii) a Nozzle meter, and (iv) a Rotameter.
2. What are the working principles of (i) an Electromagnetic Flow meter and (ii) an Ultrasonic flow meter?
3. What is “Coefficient of discharge” (C_d), for an Orificemeter, a Venturimeter and a Nozzle meter?
4. Typically, the value of $C_d < 1$. What are the probable reasons for this?

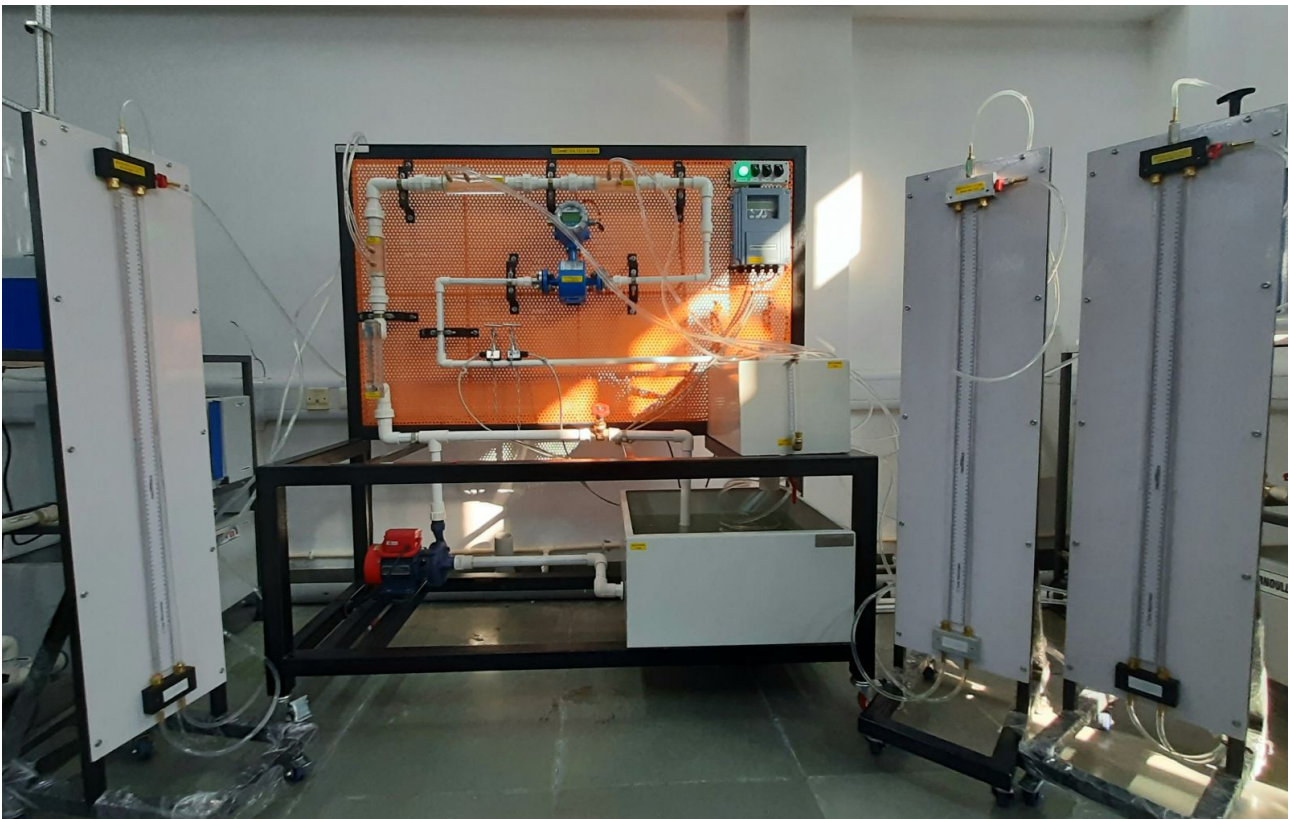


Figure 5: Photo of the experimental setup with various flowmeters.

Task 5.1: In the report for Lab Exp.-V , concise answers to all the above questions have to be submitted.

About the experiment

The experimental set up of flow meters is relatively simple (see fig. 5). A total of 6 flow meters as mentioned in items 1 and 2 of the “Essential Background” are attached in series. Therefore, the same flow passes through all of them at steady state and hence they should ideally yield the same reading. However, typically, the electromagnetic and the ultrasonic flow meters are the most accurate, while other analog flow meters produce errors with varying magnitude. Therefore, the readings from the former two flow meters may be used to estimate the coefficient of discharge of the latter ones.

Task 5.2:

- (a) Take multiple flow measurements in all flow meters in steady state.
- (b) Compute the theoretical flow rates in the analog flow meters. Are they the same as in the Electromagnetic or the Ultrasonic flow meters?
- (c) Compute the C_d values for all analog flow meters. Repeat the experiment for each flow rate at least 3 times.
- (d) Report experimental uncertainties and quantify errors by running the experiments multiple times.

Tasks 5.1 and 5.2 should be included in the report for this experiment.